COCKPIT COOLING SYSTEM

Requirements Document

For

704th Test Group/846th Test Squadron

25 November 2024

1.0 PURPOSE

The 704th Test Group (704 TG) mission is to be the premier provider of unique test and evaluation capabilities relevant to tomorrow's warfighter through world-class people improving technical excellence, cost-effectiveness, and agility. The 846th Test Squadron (846 TS) is assigned to the 704 TG. The 846 TS's primary test capability is the Holloman High Speed Test Track (HHSTT) where they utilize rocket powered sled trains to simulate flight like environments and scenarios, to include aircraft egress and ejection testing to validate aircraft components.

Specialized testing apparatuses are used to simulate specific operational environments. The 846 TS requires a Tactical Aircraft test sled cockpit cooling system which will regulate the internal temperature of the cockpit while a specialized testing apparatus, also referred to as a "hot box", will direct and distribute hot air over the external canopy transparency.

2.0 SCOPE

The Contractor will deliver the conceptual designs for a Tactical Aircraft test sled internal cockpit cooling system that meets the following requirements, as well as provide in-person support during the checkout tests and data test:

2.1 REQUIREMENTS

2.1.1 The test sled internal cockpit cooling system must be designed to be incorporated with the Tactical Aircraft test sled.

2.1.2 The test sled internal cockpit cooling system must be designed to allow the 846 TS to procure all commercial off-the-shelf (COTS) materials in a short time frame. (I.e. Parts must be readily available and not on back-order.)

2.1.3 The test sled internal cockpit cooling system must be designed in a manner in which 846 TS can fabricate all components (if necessary), with the exception of 3D printed materials, and build the cooling system within its facilities (in-house).

2.1.4 The test sled internal cockpit cooling system must be designed so that it does not inhibit, block, or prevent the following:

2.1.4.1 Opening of the hot box

2.1.4.2 The ejection or ejection path of the seat/manikin

2.1.4.3 The seat/manikin from being recovered by the catch net

2.1.4.4 Camera views of the through-canopy ejection event

2.1.4.5 Arming the seat or sled

2.1.5 The test sled internal cockpit cooling system must maintain the following temperature ranges:

2.1.5.1 Internal cockpit cameras must be less than 120 °F

2.1.5.2 Ejection seat munitions must be less than 160 °F

2.1.5.3 Helmet must be less than 158 °F

2.1.5.4 Manikin neck must be between 62-91 °F

2.1.5.5 Manikin Data Acquisition System (DAS) must be between 32-158 °F

2.1.5.6 Manikin DAS battery must be between 32-113 °F

3.0 DOCUMENTATION AND SUPPORT

3.1 The Contractor will provide a conceptual design for the sled internal cockpit cooling system.

3.2 The Contractor will provide a conceptual design of ducting/nozzles for the cooling system.

3.3 The Contractor will provide a conceptual design for cooling the mannikin head/helmet.

3.4 The Contractor will provide integration for the conceptual design of the cooling system with the existing forebody.

3.5 The Contractor support will not exceed 355 labor hours.

4.0 MILESTONES

4.1 The Contractor shall deliver preliminary design review (PDR) no later than 100 days prior to the data test date.

4.2 The Contractor will, within budget constraints, provide support of at least 1 experienced engineer during the checkout tests of system.

5.0 DELIVERABLES

5.1 The contractor shall deliver all documentation to 846 TS/TGTPM, Attn: Lt Jesiah Israel at 1521 Test Track Road, building 1179, Holloman Air Force Base, New Mexico 88330; jesiah.israel@us.af.mil.